



Review-I Technical Defense & Panel Viva Guide

Panel Q&A; Preparation, Evaluation Rubric Alignment & Individual Member Defense

Project: AI-Powered Smart Campus Resource Management System	Faculty Coordinator / Mentor: Dr. Rashmi Mothkur (ASAC CSE)
Team Lead: Likith S (2411021061437)	Members: Vinodkumar (2411021061425), Nikhil kumar V (2411021061427), Kalyan kumar T (2411021061421)

1. Review-I Evaluation Rubric Alignment

Rubric Criterion	Exemplary Standard (85–100%)	Technical Defense Evidence
1. Problem & Scope	Problem precise, bounded, evidenced by data; scope states exclusions.	Bounded strictly to Alliance University 60-acre campus (12.845° N, 77.684° E). Addressed 28.4% ghost bookings, timetable clashes, and BESCOM commercial peak demand surcharges. Exclusions: 11 kV physical substation hardware, biometric facial surveillance.
2. Literature Review	Min. 12 sources from last 5 years, comparison matrix, defended gap.	15 peer-reviewed IEEE/Elsevier research publications (2024–2026). Identified critical gap: Existing platforms decouple room scheduling from energy management and physical IoT occupancy sensing. Our platform unifies all three in a centralized intelligent system.
3. SMART Objectives	Objectives measurable and time-bound; testable functional requirements.	6 SMART objectives: Centralized multi-role dashboard (sub-100ms latency), real-time IoT monitoring (1,000+ data points), ML demand forecasting ($R^2 > 0.92$), automated timetable optimization (0 hard clashes), energy tracking & peak-shaving (18.4% cut), and administrative audit reporting.
4. Methodology & Feasibility	Methodology justified against alternatives; stack chosen on performance.	Defended tech stack: Python, JavaScript, SQL; TensorFlow, Scikit-Learn, React.js, Flask/Django; PostgreSQL, MongoDB, AWS, Firebase; Docker, Postman. Pugh decision matrices used for component selection.
5. Technical Defence	Professional delivery; every member answers on their own work.	Rigorous division of labor across 4 specialized subsystem leads with individual technical talking points below.

2. Individual Member Defense & Technical Viva Questions

Module 1: Architecture, Multi-Role Dashboard & 3D Digital Twin — Lead: Likith S (2411021061437)

- *Why WebGL over Unreal/Unity?* Game engines require multi-GB client executable downloads or expensive pixel streaming (\$2.00+/hr). Three.js WebGL runs natively inside standard web browsers on consumer hardware, achieving 60 FPS while streaming a compressed 12.4 MB binary glTF asset.
- *How was the campus 3D model built?* Authored an automated Python script (`perfect_alliance_campus.py`) executed headlessly in Blender 4.3 LTS, generating parametric meshes for Admin Block, Central Library, ACED Engineering 7-floor block, and Hostels based on real geographic coordinates (12.845° N, 77.684° E).

Module 2: Microgrid Energy Management & Power Flow — Lead: Vinodkumar (2411021061425)

- *What is the campus microgrid capacity?* 480 kWp rooftop solar PV and 1.2 MWh Battery Energy Storage System (BESS). Implemented predictive peak-shaving dispatch: midday surplus solar charges the BESS; afternoon peak discharging cuts commercial grid draw by 18.4%, avoiding BESCOM maximum demand surcharges.
- *How does the Sankey diagram work?* Computes dynamic SVG cubic Bezier paths sized proportionally to live kilowatt flows from Grid, Solar, and Battery sources to ACED, Library, Chiller Plant, and Hostels.

Module 3: IoT Telemetry Ingestion & Sensor State Simulation — Lead: Nikhil kumar V (2411021061427)

- *How do you simulate 1,000+ IoT sensors without crashing browsers?* Telemetry states in `campusData.ts` are managed through a centralized state machine batched every 1,000 ms and hydrated directly to Three.js shader uniforms and vertex buffers, keeping UI latency under 80 ms without triggering heavy React DOM re-renders.

Module 4: Automated Timetable Optimization & Space Intelligence — Lead: Kalyan kumar T (2411021061421)

- *How does the timetable solver eliminate clashes?* Formulated as a Constraint Satisfaction Problem (CSP) solved via backtracking with Minimum Remaining Values (MRV) and forward checking. Hard constraints (room/instructor overlap) are 100% satisfied; soft transit constraints are optimized via genetic operators.
- *How does the agent detect ghost bookings?* Cross-references scheduled room bookings with live PIR/thermal occupancy feeds. If a room remains unoccupied for >15 minutes after class commencement, the slot is flagged, released for walk-in bookings, and chiller setbacks are dispatched.